GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Battery Equipment Runtime Audit Worksheet

For maintenance teams and OEM designers of battery trolleys, tugs, scrubbers, AGVs and small vehicles

Log a full shift first, then work through Steps 2 to 5 to locate the losses. Complete the runtime calculation in Step 6 before and after each fix so the gain is measured, not assumed.

Step 1 — Log the duty cycle and energy used

- [ ] Battery type, nominal voltage and rated capacity: _____ V, _____ Ah at the _____ hour rate
- [ ] Fit a shunt-based energy meter or data logger at the battery terminals
- [ ] Energy and charge used per shift: _____ Wh, _____ Ah
- [ ] Average and peak battery current: _____ A average, _____ A peak
- [ ] Minimum battery voltage under load and where it occurs (ramp, start, full load): _____ V
- [ ] Ambient and battery-compartment temperature: _____ °C

Step 2 — Check the battery

- [ ] Capacity test at the rated discharge rate; plan replacement if below about 80 % of rated capacity
- [ ] Peukert-corrected capacity: C eff = C × (C / (I × H)) raised to the power (k - 1)
- [ ] Use k of about 1.1–1.3 for lead-acid (higher for older flooded cells) and about 1.0–1.05 for lithium-ion
- [ ] Usable depth of discharge: typically 50–80 % for lead-acid, 80–90 % or more for LiFePO4
- [ ] Charging regime: full charge after each shift, correct charge voltage, equalising where the maker specifies it

Step 3 — Measure electrical losses

- [ ] Voltage drop from battery posts to motor terminals at rated current: _____ V (investigate above about 2 % of nominal)
- [ ] Millivolt drop across each fuse, contactor, connector and crimp under load
- [ ] Check cables and lugs by touch or thermal camera after a heavy run; warm joints indicate high resistance
- [ ] Controller: confirm PWM frequency and current limit suit the motor; typical efficiency 95–98 %

Step 4 — Check the motor operating point

- [ ] Cruise current as a percentage of motor stall current: _____ % (PMDC peak efficiency is typically near 10 %)
- [ ] Cruise current well above 25 % of stall: the motor is working hard; consider re-gearing or a larger frame
- [ ] Cruise current only a few per cent of stall: the motor is oversized for the duty
- [ ] Motor runs near rated speed at the most-used travel speed; long periods at low PWM duty waste energy
- [ ] Brushes and commutator inspected: worn brushes and poor film raise contact drop

Step 5 — Remove mechanical drag

- [ ] Towing test with brakes released: rolling resistance coefficient = towing force / (mass × 9.81)
- [ ] Tyre pressures to specification; no flat-spotted or damaged tyres
- [ ] Electromagnetic brake air gap within specification and no drag when released
- [ ] Chains and belts aligned and tensioned; gearbox lubricant at the correct level and grade

- [ ] Wheel and axle bearings turn freely by hand

Step 6 — Calculate runtime

- [ ] System efficiency = wiring × controller × motor × gearbox efficiencies
- [ ] Battery power = average mechanical power / system efficiency
- [ ] Battery current = battery power / nominal voltage
- [ ] Runtime (h) = usable Ah at that current (Peukert-corrected × depth of discharge) / battery current
- [ ] Repeat after each fix and record the gain

Typical Efficiency Ranges for Planning

Stage	Typical efficiency	Main loss mechanism
Cables and connectors	97–99 %	I^2R in conductors and joints
PWM controller	95–98 %	Switching and conduction losses
PMDC motor at its best point	78–88 %	Copper, brush, iron and friction losses
BLDC motor at its best point	85–95 %	Copper and iron losses
Helical or spur gear stage	95–98 % per stage	Tooth friction and churning
Worm gear stage	40–90 %, falling with ratio	Sliding friction
Lead-acid battery at high current	Peukert k of 1.1–1.3	Reduced deliverable capacity

Ranges are typical planning values; use measured data wherever possible. Send the completed worksheet to Globe Scott Motors for a review of motor sizing and drivetrain options.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.